

Technical Design Document

Programming Exercise 2: Spam Message Checker

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Program Purpose

The purpose of this program is to analyze an email message and determine the likelihood that it is spam. The program checks the message for common spam words and phrases. Each spam word or phrase found increases the spam score by one point. After checking the message, the program displays the spam score, the likelihood of spam, and the spam words or phrases that were found.

Input

The program accepts the following input:

- An email message entered by the user.

Output

The program displays:

- The spam score.
- The likelihood of the message being spam (Low, Medium, or High).
- A list of spam words or phrases found in the message.

Functions

`get_message()`

Purpose:

- Prompts the user to enter an email message.
- Returns the message entered by the user.

Parameters:

- None

Returns:

- Email message (string)

`check_spam(message, spam_words)`

Purpose:

- Searches the email message for spam words and phrases.
- Stores any matching words or phrases in a list.

Parameters:

- message (string)
- spam_words (list)

Returns:

- List of detected spam words and phrases

`get_likelihood(score)`

Purpose:

- Determines the likelihood that a message is spam based on the spam score.

Parameters:

- score (integer)

Returns:

- Spam likelihood (string)

`display_results(score, likelihood, found_words)`

Purpose:

- Displays the spam score.
- Displays the likelihood of spam.
- Displays the spam words or phrases found.

Parameters:

- score (integer)
- likelihood (string)
- found_words (list)

Returns:

- None

Processing Logic

Create a list containing 30 common spam words and phrases.

Prompt the user to enter an email message.

1. Convert the message to lowercase for comparison.
2. Check the message against each spam word or phrase.
3. Store matching words or phrases in a list.
4. Calculate the spam score using the number of matches found.
5. Determine the likelihood of spam based on the score:
 - o 0-2 = Low
 - o 3-5 = Medium
 - o 6 or more = High
6. Display the spam score.
7. Display the likelihood of spam.
8. Display the spam words or phrases found.

Variable	Type	Purpose
message	String	Stores the email message entered by the user
spam_words	List	Stores the 30 spam words and phrases
found_words	List	Stores detected spam words and phrases
score	Integer	Stores the spam score
likelihood	String	Stores the spam likelihood rating

OUTPUT IN PHOTO:

Enter an email message:

"Congratulations! Click here to claim your free prize."

Output:

Spam Score: 5

Likelihood: Medium likelihood of spam

Words/Phrases Found:

- free
- click here
- congratulations
- claim
- prize

```
*C:\Users\The Computer Place\PyCharmMiscProject\.venv\Scripts\python.exe" *C:\Users\The Computer Place\PyCharmMiscProject\Programming_Exercise_2.py"
Enter an email message:
Congratulations! Click here to claim your free prize.

Spam Score: 5
Likelihood: Medium likelihood of spam

Words/Phrases Found:
- free
- click here
- congratulations
- claim
- prize

Process finished with exit code 0
```